

2024



DSM Accelerates Development to Deliver 150+ Apps Using Mendix Low-Code Software on AWS

The IT team of dsm-firmenich is always looking for new ways to use technology to deliver solutions across the organization. It wanted to modernize and standardize its applications across the global company to create high-quality solutions at speed. “Anything that cannot be done with a standard application, we build ourselves,” says Anouke Tielens-Coenegracht, the dsm-firmenich chapter lead for technology development. “Things are moving to the cloud more and more.”

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

5
times | faster app development

50
different countries | being supported



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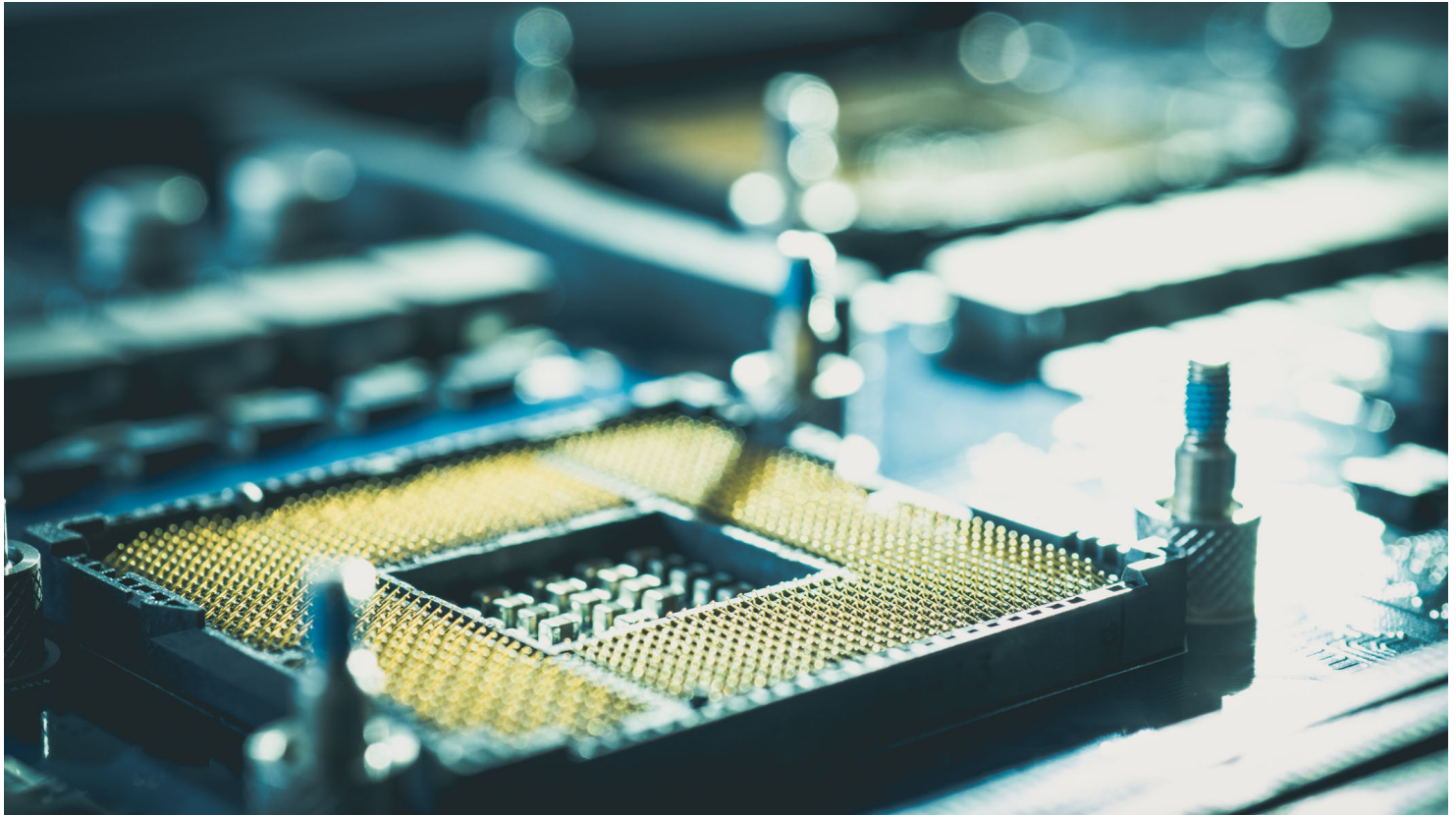
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Overview

Health and nutrition leader dsm-firmenich uses AWS and the Mendix low-code platform to create custom-build applications.

Since 2018, DSM, now part of dsm-firmenich, had been using Amazon Web Services (AWS) as its main cloud provider. The company wanted to make it even simpler for teams to develop applications to address various use cases. DSM turned to Mendix, a subsidiary of Siemens Digital Industries Software, an AWS Partner. Using Mendix, DSM teams increased the speed of app development by five times and automated time-consuming manual processes with about 150 applications for internal teams and customers.



Modernizing Legacy Projects through Mendix Low-Code Technology

An initialism for Dutch State Mining, DSM formed in 1902 as a state-owned coal-mining company. The company evolved its focus from coal mining to petrochemicals to nutrition, health, and sustainable living. A merger with the Swiss company Firmenich in May 2023 created dsm-firmenich, a world leader in nutrition, health, and beauty, with revenue exceeding 12 billion dollars and 30,000 employees globally. The company operates in about 50 countries using disparate solutions that include a large SAP implementation, on-premises workflows, and applications running in the cloud. Looking for a way to build tools that connect these solutions, in 2016, legacy DSM turned to Mendix, a low-code software development platform that is intertwined and highly compatible with AWS. Rather than rely on programming languages, low-code software provides a visual interface that simplifies building web and mobile applications for developers and nontechnical users.

Many internal legacy DSM teams were using data from spreadsheets or siloed SQL databases. Legacy DSM began to use Mendix as a way to modernize legacy projects. “People were using c



spread all over,” says Tielens-Coenegracht. “Using Mendix, we are in control from an IT perspective. We can help business areas to build an application. They don’t have to go shopping for different vendors or different types of solutions.”

Simplifying App Creation to Automate Processes for Customers

On average, one developer can create an app in four or five 2-week sprints, using technology that the company implements to simplify the process. dsm-firmenich created a starter app that helps internal teams find dsm-firmenich default branding to standardize the look of apps created on Mendix. A developer can download app components from the Mendix marketplace or from the company’s private store. “Mendix created a complete package where everything comes together in such a nice way,” says Wouter Vijverberg, dsm-firmenich’s lead technology development expert. “Anyone can create an account and have the freedom to learn. The productivity and speed of development work well.”

To connect its workflows, dsm-firmenich uses Mendix to consume APIs that are custom built in a middleware solution, creating a repository of APIs that teams can reuse for many SAP data objects. In fact, half of the Mendix applications interact with SAP for pricing and quotations, mobile warehouse functions and maintenance, customer portals, and other functions. “We want to make our processes more efficient so that people have more time to spend on other topics,” says Tielens-Coenegracht.

For example, part of its app portfolio is a customer-invoice portal on Mendix that automatically manages purchase orders. When a customer emails a purchase order, the app uses optical character recognition and machine learning to automatically extract details and create a corresponding sales order in SAP. The customer service team can follow the status of the sales order from purchase through payment. “The customer doesn’t have to do anything if the data is right, and if not, customer service employees can simply change and resubmit the data,” says Vijverberg. “We are constantly improving the workflow with custom rules so that it performs even better. The aim is to make the process almost touchless.”

Incorporating Mendix and AWS to Drive Business Val



Mendix is cloud native, so by default dsm-firmenich doesn't require additional solutions experts to deploy to [Amazon Virtual Private Cloud \(Amazon VPC\)](#), which lets companies define and launch AWS resources in a logically isolated virtual network. The company built a continuous integration and delivery pipeline to deploy around 150 Mendix applications to Amazon VPC. "Having deployment completely automated is much simpler than doing it manually," says Vijverberg. "That's also why we grew so fast. It's simple to deploy multiple applications, and we can spin up a new Mendix app in minutes."

Using Mendix, the teams develop applications five times faster than they do using traditional SAP methodologies. "We try to embrace AWS where we can," says Vijverberg. "If Mendix does not provide the wanted functionality or capability, we use AWS services to bridge that gap." For example, dsm-firmenich created a video-learning app for QR code labels on machines in production. After the app scans a QR code, the app plays back recorded instructions that it accesses from [Amazon Simple Storage Service \(Amazon S3\)](#), an object storage service offering industry-leading scalability, data availability, security, and performance. In another application, users access a URL to play video stored in Amazon S3 on a Mendix video player.

Additionally, when a user uploads a video to a bucket in Amazon S3, the upload automatically generates thumbnail images using [AWS Lambda](#), a serverless, event-driven compute service that lets developers run code for virtually any type of application or backend service without provisioning or managing servers. "We have a lot of low-code applications running in Mendix," says Vijverberg. "That means that it's simpler to maintain and less expensive over the life cycle."

Standardizing Technology for the Future Using AWS and Mendix

Because legacy Firmenich did not use a low-code platform, the teams expect to use Mendix as the low-code platform of choice. They anticipate demand for Mendix applications to increase, creating many additional opportunities as dsm-firmenich reorganizes and consolidates its digital and technology organization.

The Mendix platform is open at every level, and Mendix apps are extensible with reusable components and custom code. As the foundation of its cloud operations, AWS offers high reliability for the performance of Mendix apps. "We knew if you want to be scalable and build solutions that really work, it is better in the end to go for your own infrastructure," says Vijverberg. "We chose Mendix and AWS as our innovation solutions for a reason."

About DSM

From century-old roots as the Dutch State Mines, DSM has evolved into a purpose-led science-based global company specializing in solutions for human and animal health and nutrition. DSM's purpose is to create brighter lives for all.

AWS Services Used

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

AWS Virtual Private Cloud

Amazon Virtual Private Cloud (Amazon VPC) gives you full control over your virtual networking environment, including resource placement, connectivity, and security.

[Learn more »](#)

AWS S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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JAPAN



Hitachi on AWS

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2021





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2023



KABAM Robotics enhances
robot intelligence with...

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